Chap 10: Lists

CS 101- Programming Fundamentals

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List

- Sequence of values (called **elements** or **items**)
- Values can be of **any type**:
  - [1 , 2 , 6 , 16]
  - [CS, EE, CnD, SDP]
  - ['spam', 2.0, 5, [10, 20]]
- A list within another list is **nested.**
- list that contains no elements is called an **empty list.**

List (cont.)

- Lists are mutable

```
marks= [25, 25, 30, 35]

marks[0] = 30

print (marks)
```

List Traversal

- For loop: to read the elements of the list.
- For loop with *range* and *len:* to write or update the elements

```
for i in range(len(numbers)):
    numbers[i] = numbers[i] * 2
```

- A for loop over an empty list never runs the body

List operations

- Concatenation (+):

```
>>> a = [1, 2, 3]
>>> b = [4, 5, 6]
>>> c = a + b
>>> c
[1, 2, 3, 4, 5, 6]
```

- Repetition (*):

```
>>> [0] * 4
[0, 0, 0, 0]
>>> [1, 2, 3] * 3
[1, 2, 3, 1, 2, 3, 1, 2, 3]
```

List methods

- **Append:** Adds a new element to the end of a list
- **Extend:** takes a list as an argument and appends all the elements
- **Sort**: Arranges the elements of the list from low to high
- **Insert:** Inserts an item at the defined index
- **Remove:** Removes an item from the list
- **Pop:** Returns and removes an element at the given index

- Deleting elements/list

```
# Deleting list items
marks= [25, 30, 30, 22, 35, 20]

# delete one item
del marks[3]

# delete multiple items
del marks[1:4]

# delete the entire list
del marks
```

Map, filter, reduce

- Reduce: Combining a sequence of elements into a single value
- Map: Map a function onto each of the elements in a sequence.
- Filter: Select some of the elements and filter out the others.

Nested lists

- A list that is an element of another list.

  t = [[1, 2], [3], [4, 5, 6]]

- Accessing nested list elements.
- Modifying nested list elements.
- Traversing nested list.

Nested list operations

- Nested list length
- Inserting an item
- Removing an item
- Sorting the list
- Append an item to nested list

Adding, multiplying and transposing matrices

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} + \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+9 & 2+8 & 3+7 \\ 4+6 & 5+5 & 6+4 \\ 7+3 & 8+2 & 9+1 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$